Neural Surrogates of Compound Flooding That Transfer to Unseen Tropical Cyclones with One Simulation

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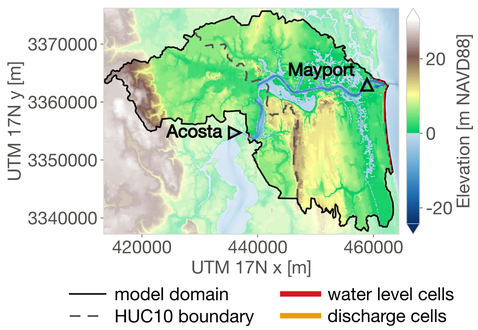
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Civil & Environmental Engineering
Center for Engineering Resilience and Climate Adaptation

Introduction & Motivation

- Tropical cyclones drive compound coastal flooding from storm surge, rainfall and river discharge in Jacksonville.
- Physics-based models such as SFINCS are too slow for real-time forecasts and large ensembles.
- Neural surrogates are fast, but observed data from past extremes are limited, so how well do they handle an unseen tropical cyclone?

Methods

Physical model setup

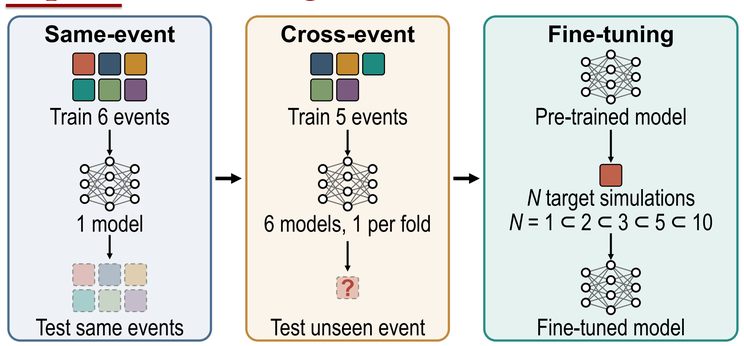


Study area on the lower St. Johns River in Jacksonville, FL

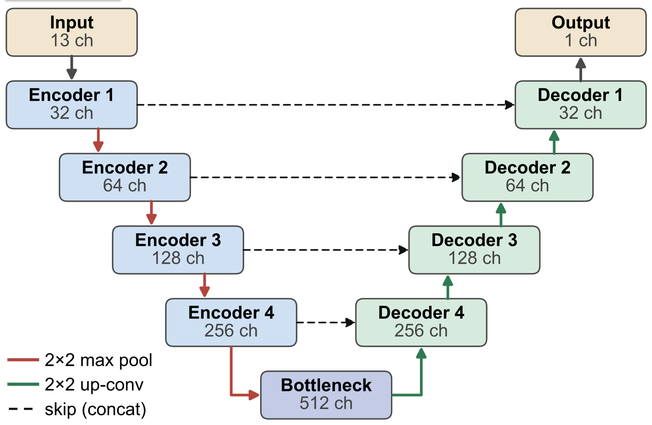
Tropical cyclone
Beryl 2012
Matthew 2016
Irma 2017
Dorian 2019
Ian 2022
Milton 2024

- SFINCS is forced by rainfall, river discharge and water level for 10 days, a 3-day spin-up + 7-day event period.
- 6 events × 81 forcing variants = 486 SFINCS simulations
- Train / validation / test = 60 / 11 / 10 simulations per event

Experimental design

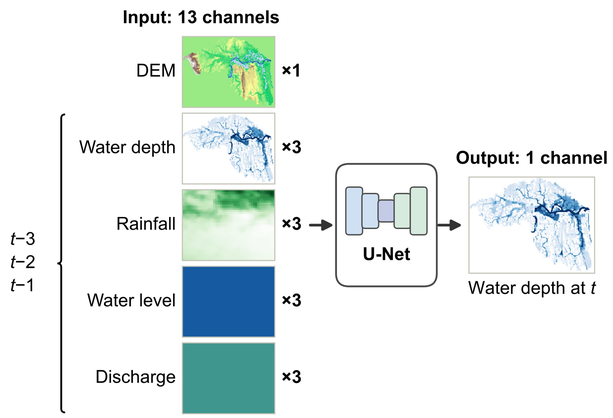


Surrogate model setup



U-Net architecture

The U-Net learns global patterns and local details.

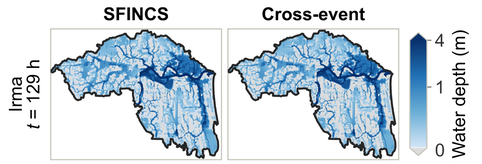


Surrogate model workflow

Image-to-image prediction: 13 inputs → 1 map

Results

Accuracy on seen and unseen events



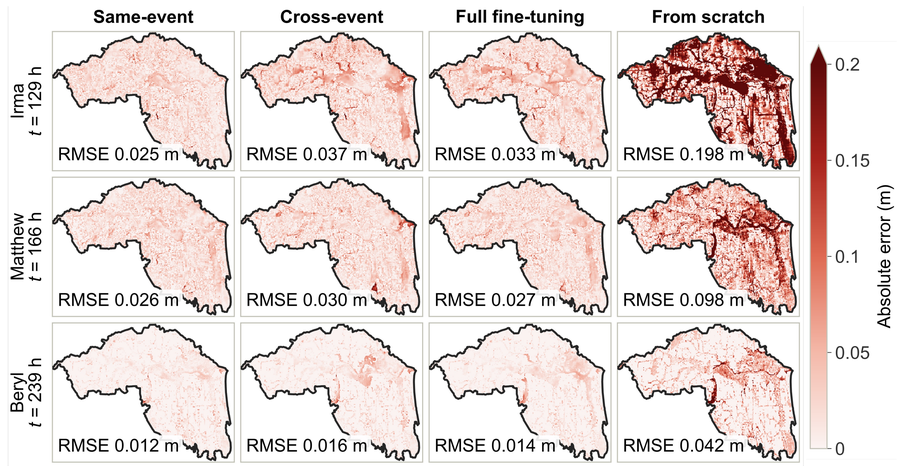
Water depth at the peak flood hour

Model	RMSE
Same-event	0.0178 m
Cross-event	0.0204 m
Persistence	0.1017 m

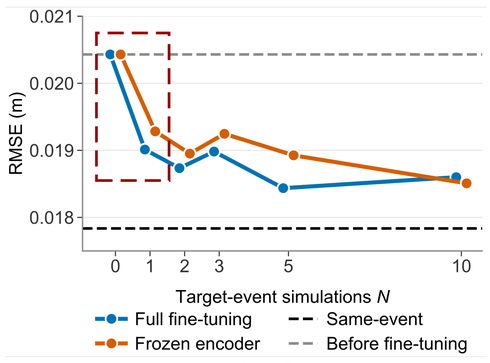
+15%

- RMSE on unseen events rises only **15%**, far below persistence.
- The surrogate runs **18×** faster than SFINCS, enabling larger ensembles.

Fine-tuning with few target simulations



Absolute error at the peak flood hour



6-fold mean RMSE after fine-tuning

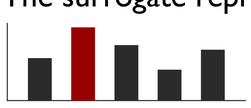
- **1** target simulation recovers most of the loss, and more simulations bring little further improvement.
- Updating only the decoder performs close to full fine-tuning.
- Pre-training is essential, as scratch training on the same data fails.

Conclusions & Future Work

Conclusions

The surrogate reproduces SFINCS compound-flood depth maps, generalizes well to unseen tropical cyclones, and adapts to new events with just **1** target simulation.

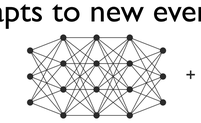
Future work



Model benchmarks
FNO, GNN, Transformers



Broader transfer
Watersheds, resolutions



AI with physics
PINNs, mass/momentum conservation